

Smart and Connected Cities

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Background

- One-third of the world's population lived in cities in 1950; by 2050, two-thirds of our planet's population will be city dwellers.
- Africa and Asia, which today account for 90% of the world's rural population, are projected to have 56% and 64%, respectively, of their populations urbanize
- In terms of raw numbers, the urban population of the world has grown to nearly 4 billion, from just 746 million in 1950. By 2050, this figure will grow by another 2.5 billion.
- Today, cities are responsible for 60% to 80% of the world's energy and greenhouse emissions and consume 60% of all potable water, losing as much as 20% in leakage

- A recent Cisco study, as illustrated in Figure 12-1, expects IoT to have the following economic impact over a 10-year period:³
- **Smart buildings:** Smart buildings have the potential to save \$100 billion by lowering operating costs by reducing energy consumption through the efficient integration of heating, ventilation, and air-conditioning (HVAC) and other building infrastructure systems.
- **Gas monitoring:** Monitoring gas could save \$69 billion by reducing meter-reading costs and increasing the accuracy of readings for citizens and municipal utility agencies.
- **Smart parking:** Smart parking could create \$41 billion by providing real-time visibility into parking space availability across a city. Residents can identify and reserve the closest available space, traffic wardens can identify noncompliant usage, and municipalities can introduce demand-based pricing.
- **Water management:** Smart water management could save \$39 billion by connecting household water meters over an IP network to provide remote usage and status information.
- **Road pricing:** Cities could create \$18 billion in new revenues by implementing automatic payments as vehicles enter busy city zones while improving overall traffic conditions

Who Benefits?

By enabling new and more meaningful connections, governments and other public-sector agencies worldwide can benefit and ultimately create quantifiable benefits for citizens.

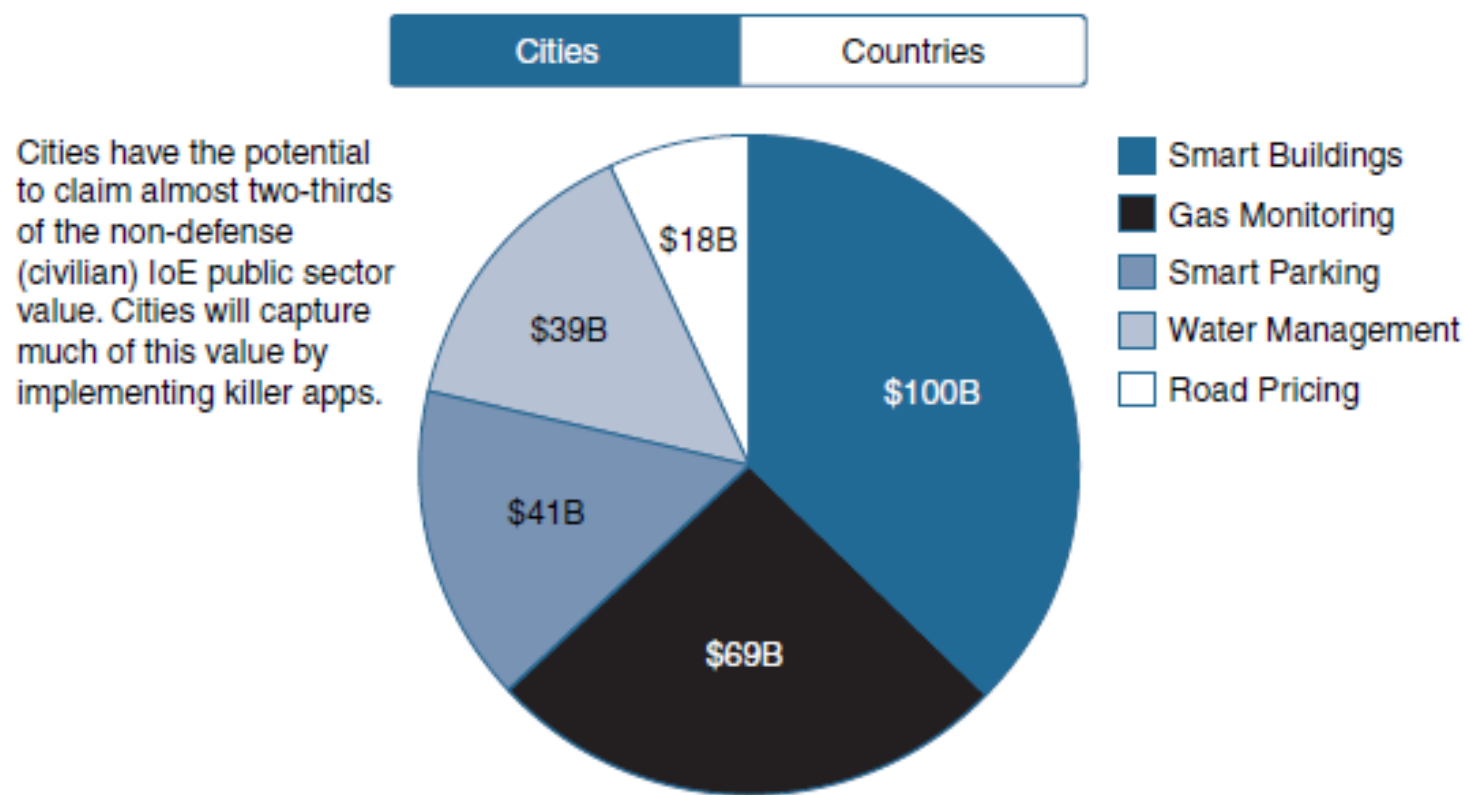


Figure 12-1 *Key Use Cases for Smart Cities*

Smart City IoT Architecture

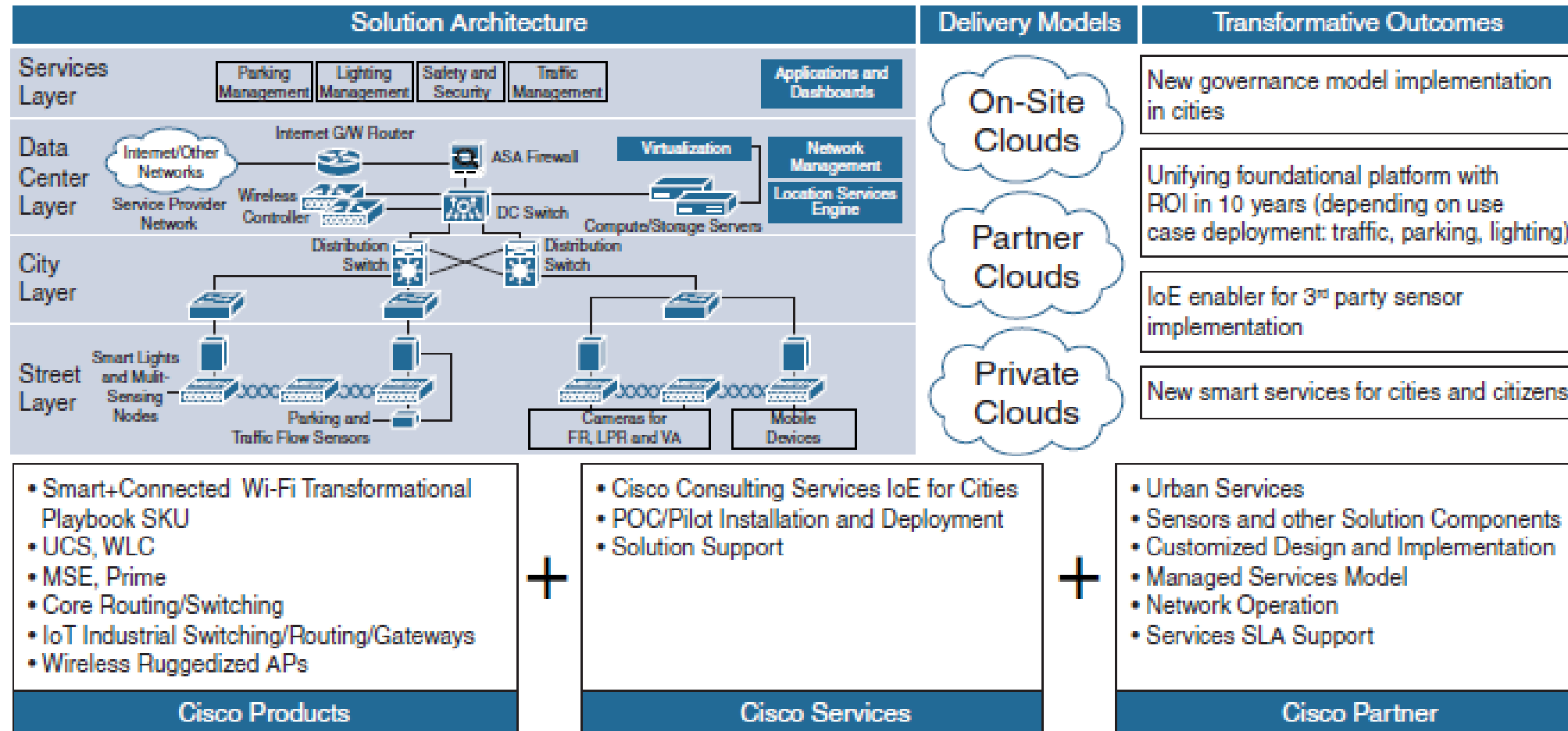


Figure 12-2 *Smart Cities Layered Architecture*

Smart City IoT Architecture

- **Street Layer:** The street layer is composed of devices and sensors that collect data and take action based on instructions from the overall solution, as well as the networking components needed to aggregate and collect data

A variety of sensors are used at the street layer for a variety of smart city use cases. Here is a short representative list:

- A **magnetic sensor** can detect a parking event by analyzing changes in the surrounding magnetic field when a heavy metal object, such as a car or a truck, comes close to it (or on top of it).
- A **lighting controller** can dim and brighten a light based on a combination of time-based and ambient conditions.
- **Video cameras** combined with video analytics can detect vehicles, faces, and traffic conditions for various traffic and security use cases.
- An **air quality sensor** can detect and measure gas and particulate matter concentrations to give a hyper-localized perspective on pollution in a given area.
- **Device counters** give an estimate of the number of devices in the area, which provides a rough idea of the number of vehicles moving or parked in a street or a public parking area, of pedestrians on a sidewalk, or even of birds in public parks or on public monuments—for cities where bird control has become an issue

- **City Layer:** At the city layer, which is above the street layer, network routers and switches must be deployed to match the size of city data that needs to be transported. This layer aggregates all data collected by sensors and the end-node network into a single transport network.
- one key consideration of the city layer is that it needs to transport multiple types of protocols, for multiple types of IoT applications.
- Some applications are delay- and jitter-sensitive, and some other applications require a deterministic approach to frame delivery.
- the city layer must be built around resiliency, to ensure that a packet coming from a sensor or a gateway will always be forwarded successfully to the headend station

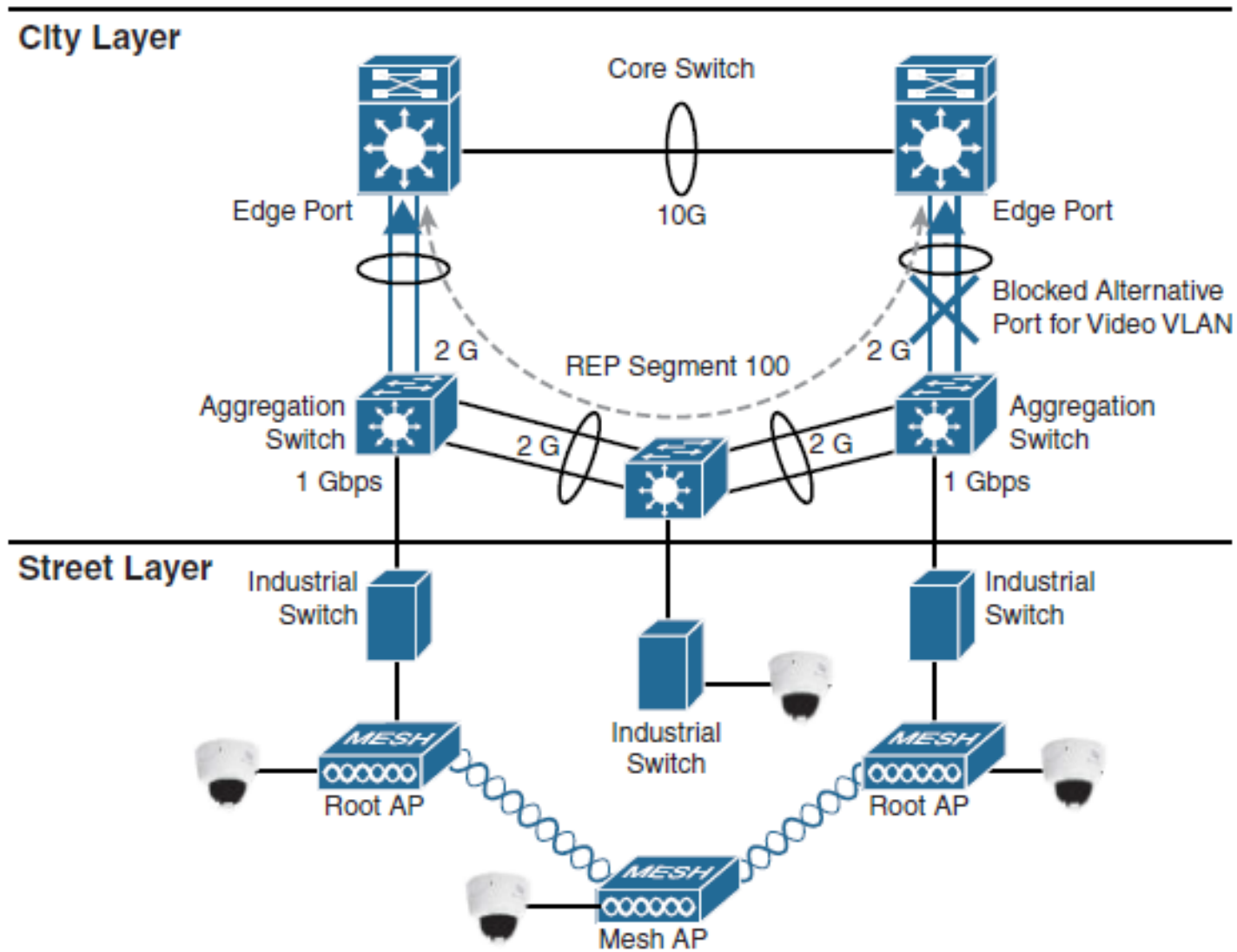


Figure 12-3 *Street Layer Resiliency*

- **Data Center Layer:** Ultimately, data collected from the sensors is sent to a data center, where it can be processed and correlated. Based on this processing of data, meaningful information and trends can be derived, and information can be provided back

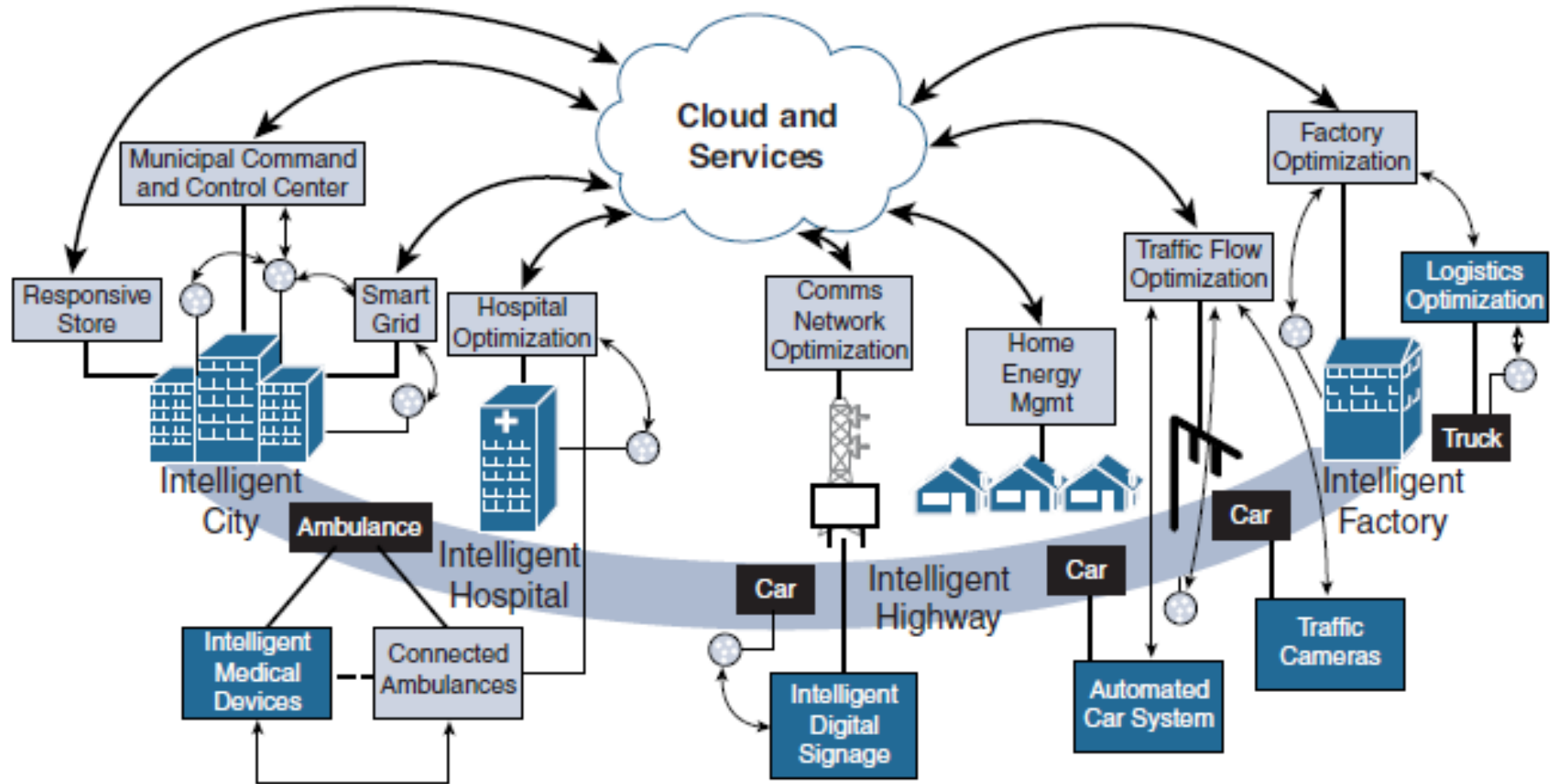


Figure 12-4 *The Role of the Cloud for Smart City Applications*

- Services Layer: Ultimately, the true value of ICT connectivity comes from the services that the measured data can provide to different users operating within a city
- Examples:
 - Parking data
 - Traffic Information
 - Street Lighting
 - Waste Management

Smart City Use Case Examples

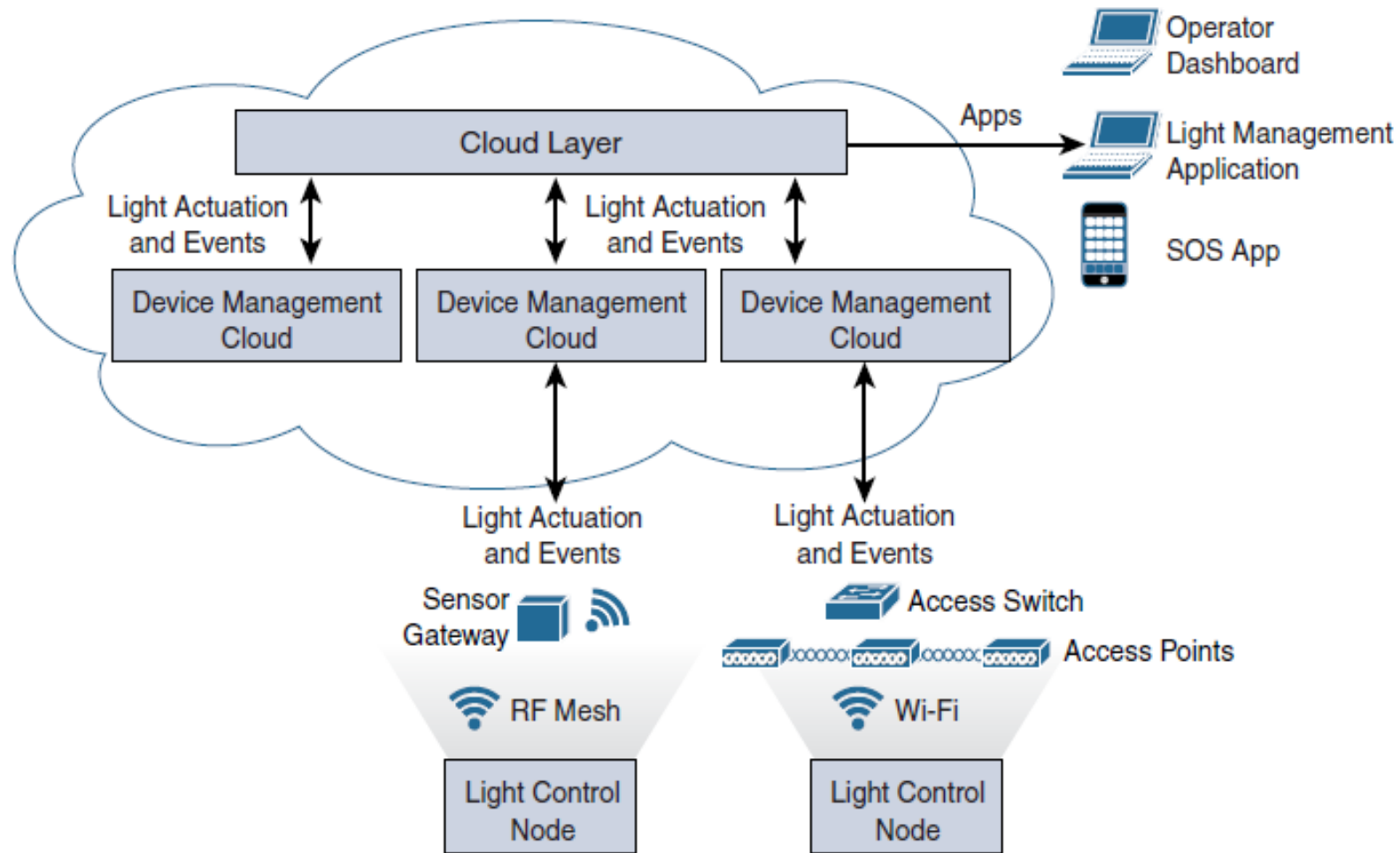


Figure 12-7 *Connected Lighting Architecture*

- A human or automated operator can use a cloud application to perform automated scheduling for lights and even get light sensors to perform automated dimming or brightening, as needed
- Lighting nodes vary widely in the industry, especially with respect to elements such as what communication protocol they use (for example, Wi-Fi, cellular, ZigBee, 802.15.4g [Wi-SUN], LoRaWAN)
- LED light bulbs are commonly equipped with basic sensors that can detect light (driving local on/off and dimming actions) and that can also detect many other environmental parameters, such as temperature, motion, pressure, or humidity. Street lights can also become local weather reporting stations

Smart Parking

Smart+Connected Parking High-Level Architecture

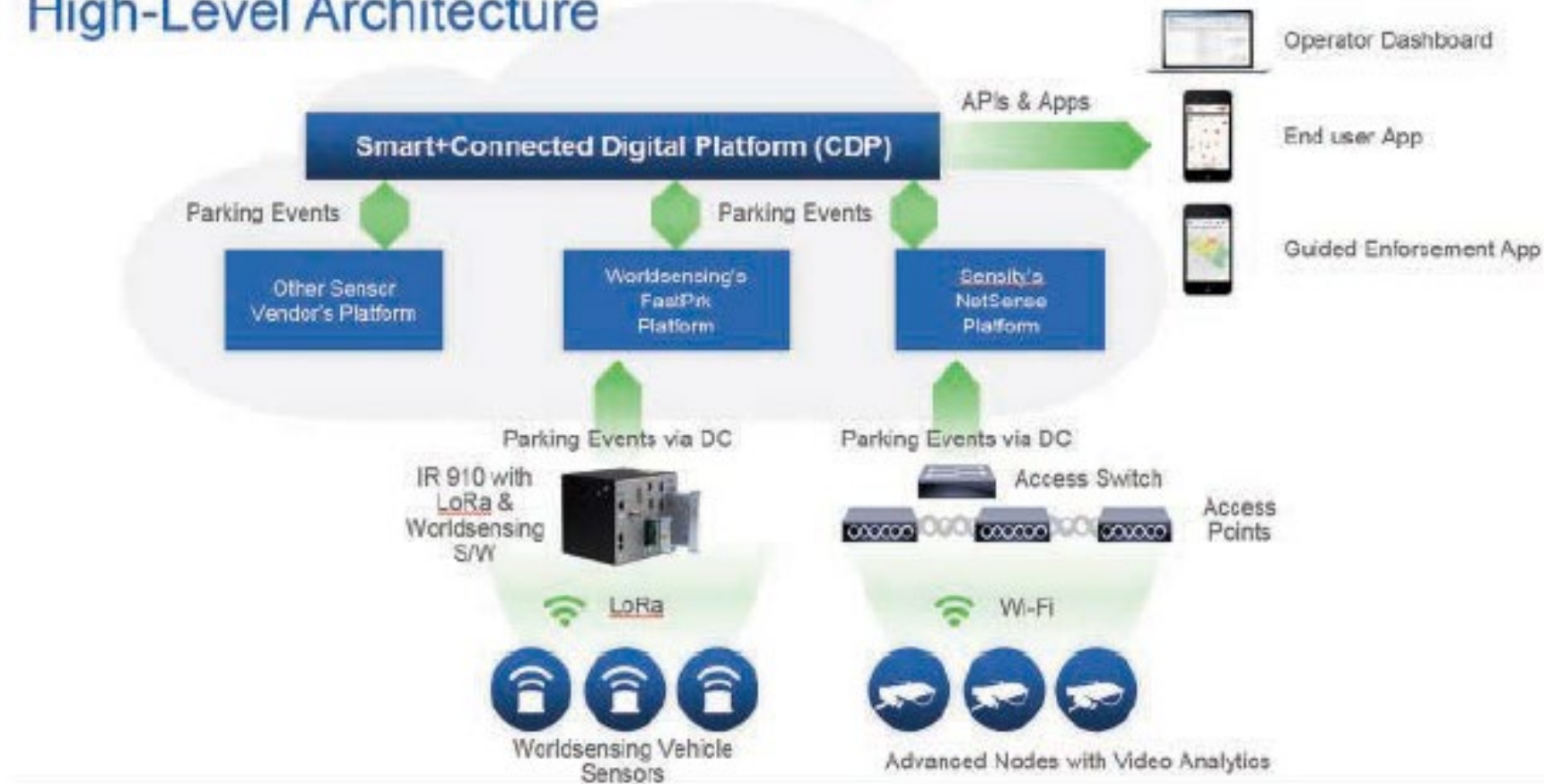


Figure 12-8 *Connected Parking Architecture*

Smart Traffic Control Architecture

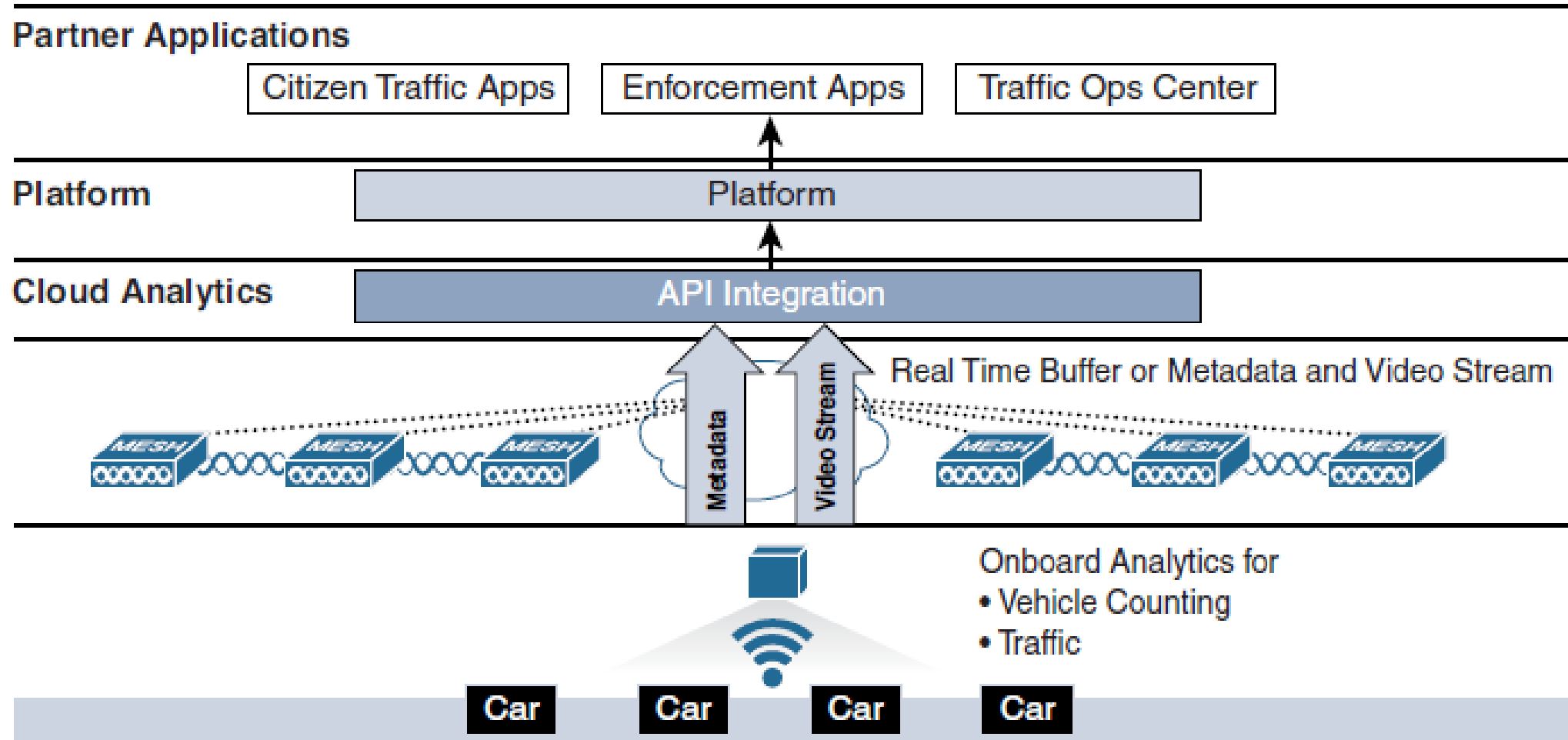


Figure 12-9 Smart City Traffic Architecture

Connected Environment Architecture

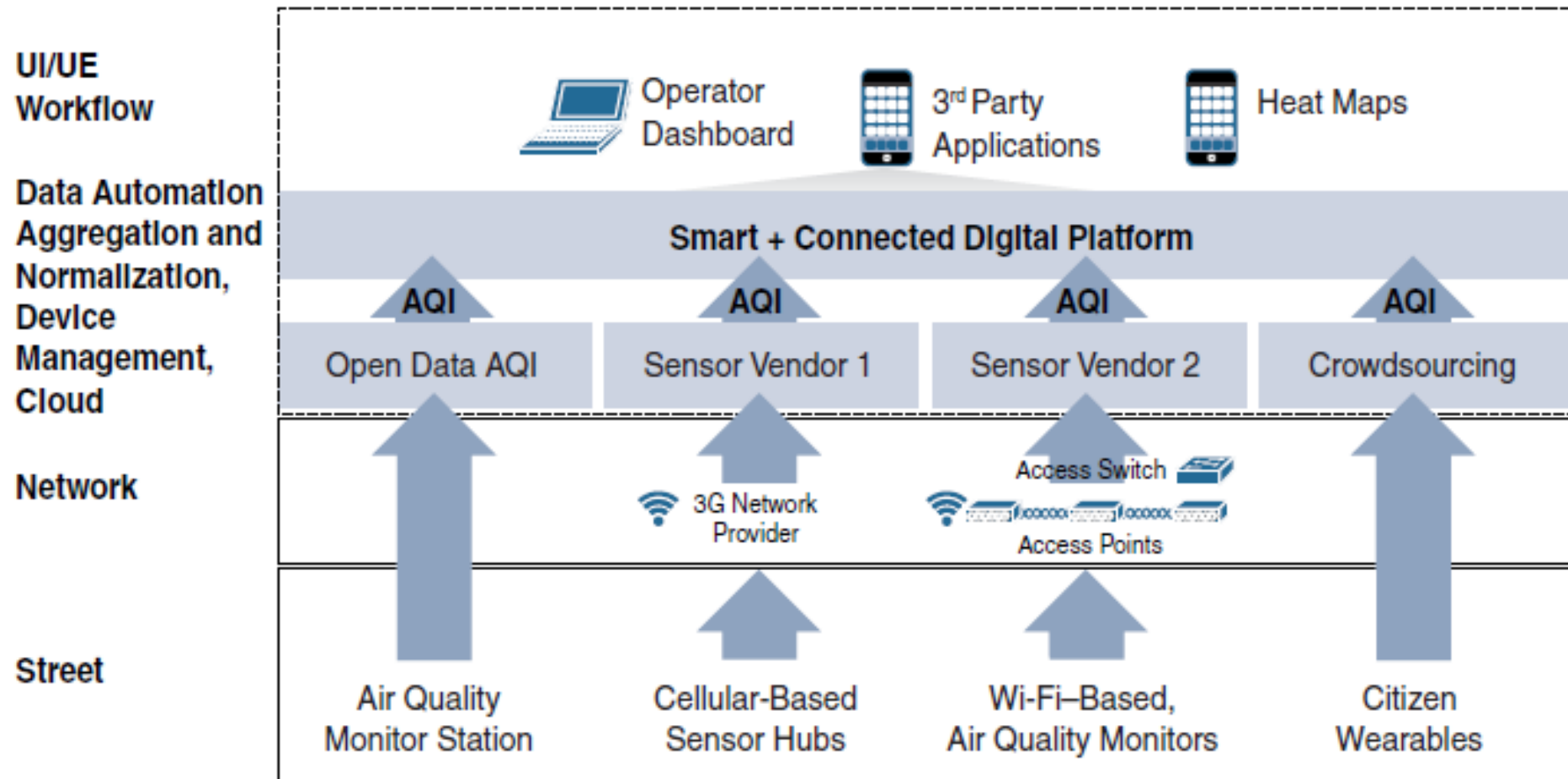


Figure 12-10 *Connected Environment Architecture*